SHAIPED Deliverable 3.7: Demonstrator showcasing data access to augment EHDS dataset catalogues with Generative AI

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Abstract

Building upon the HealthDCAT-AP RDF vocabulary, the SHAIPED D3.7 demonstrator showcases an augmented dataset catalogue that leverages Generative AI (GenAI) to enhance dataset discovery. Within the context of the European Health Data Space (EHDS), the demonstrator translates research questions into semantically coherent dataset representations through a Virtual Dataset Builder. Using retrieval-augmented generation (RAG) and large language model embeddings, the system identifies relevant dataset variables, constructs conceptual "virtual datasets," and proposes synthetic proxy data to support privacy-preserving exploration and early-stage prototyping analyses. This approach provides a human-friendly layer on top of semantic web technologies such as DCAT, CSVW, and Wikidata, bridging the gap between sensitive health data and research applications.

Keywords

Generative AI, FAIR data, HealthDCAT-AP, synthetic data, dataset catalogue, European Health Data Space (EHDS), retrieval-augmented generation (RAG)

1. Introduction

The European Health Data Space (EHDS)[1] aims to improve access to health data for research, innovation, and policy-making while ensuring privacy and security. In this context, the HealthDCAT-AP[2] standard provides a unified semantic framework for describing health datasets through machine-readable, actionable metadata, enabling consistent dataset cataloguing and discovery across stakeholders. The EU funded project SHAIPED[3] supports Health Data Access Bodies (HDABs) by creating AI-ready pathways that accelerate the development of AI as medical device tools. It enhances access to health data, streamlines data workflows for AI applications, and strengthens the EHDS ecosystem.

SHAIPED Deliverable 3.7 introduces a Generative AI-driven demonstrator that:

- Uses synthetic proxy data to mimic sensitive health datasets, enabling safe exploration.
- Implements a semantic layer (HealthDCAT-AP[4] [1], CSVW[5], Wikidata[6]) to standardise dataset descriptions.

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[1] Regulation (EU) 2025/327 of the European Parliament and of the Council of 11 February 2025 on the European Health Data Space and amending Directive 2011/24/EU and Regulation (EU) 2024/2847". http://data.europa.eu/eli/reg/2025/327/oj

[2] Health Data Cataloging Literacy with HealthDCAT-AP. https://metadata.healthdataportal.eu/

[3] SHAIPED's overarching goals include the establishment of a streamlined, regulatory-aligned framework for AI medical devices, centered around the main goals. https://shaiped.eu

[4] Derycke P. et al. HealthData@EU Pilot – Deliverable 6.2 HealthDCAT-AP. Zenodo DOI 10.5281/zenodo.15085025.

[5] https://www.w3.org/ns/csvw

[6] https://www.wikidata.org

- Leverages BioMistral (a generative AI model)[7] through retrieval-augmented generation (RAG)[8] to construct context-aware responses to research queries, mapping them to real-world datasets.

This approach enhances data discoverability without exposing sensitive data, highlighting the demonstrator's potential to advance data-driven research and innovation within the EHDS.

2. Use case and motivation

The demonstrator builds on HealthDCAT-AP to enhance dataset discovery while safely exposing synthetic proxy data. It adds a human-friendly layer on top of semantic standards such as DCAT[9], CSVW, and Wikidata, helping users navigate machine-readable metadata without requiring deep semantic-web expertise.

When a user submits a research question, a generative AI model constructs a coherent and context-aware response tailored to the user's research question that includes:

- a conceptual virtual dataset (a view of relevant variables and relationships);
- a mapping to real-world datasets that can be made available for analysis, for example via an EHDS access request;
- a proposal for synthetic data generation to guide downstream analysis.

By enabling early prototyping on synthetic data and a clear conceptual model, researchers can explore feasibility and refine analysis plans without immediate access to sensitive health data. This reduces barriers to data exploration and supports FAIR-aligned metadata reuse and interoperability [2].

3. Discussion

This approach illustrates how Generative AI can bridge semantic data standards and human-centric discovery interfaces. While synthetic data cannot replace sensitive health data, it serves as a proxy for query validation and workflow testing. Future work will investigate how to integrate federated analysis and privacy-preserving data visiting principle (PHIRI[10]) to further align with the EHDS Regulation.

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Declaration on Generative AI

During the preparation of this work, the author(s) used CHAT-GPT-5 in order to: Grammar and spelling check. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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